T25(iv): Conditional evidence variance under PW2

SMC-BVS v3 proof run

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Scope. This document proves only the conditional assertion in T25(iv), including four-island pooling. It does not establish PW2 for unrestricted inputs. All notation and hypotheses are those of `STATEMENT.md`. The proof is not an adoption of the result.

1 Setting and precise conclusion

Fix the data and a successful frozen calibration value c. The finite state space is $\Omega = \{S \subseteq [p] : |S| \le K\}$, with $K = \min(p, \lfloor n/2 \rfloor)$. The method's positive bridge is $\gamma_\varphi = \gamma_0 e^{\varphi d}$, with $\bar{Z}_\varphi = \sum_{S\in\Omega} \gamma_\varphi(S)$ and $q_\varphi = \gamma_\varphi/\bar{Z}_\varphi$. Here $\gamma_0(S) = \varpi(S)\prod_b r_b(S \cap C_b)$, $r_b(U) = m(U)/m(\emptyset)$, and $d(S) = \log[m(S)/m(\emptyset)] - \sum_b \log r_b(S \cap C_b)$. Consequently $\gamma_1(S) = \varpi(S)m(S)/m(\emptyset)$.

Write $N = N_e$ for the number of particles in one island and fix $0 = \varphi_0 < \cdots < \varphi_T = 1$. Put

$$q_t = q_{\varphi_t}, \qquad Z_t = m(\emptyset)\bar{Z}_{\varphi_t}, \qquad G_t = e^{(\varphi_t - \varphi_{t-1})d}, \qquad c_t = \frac{Z_t}{Z_{t-1}}, \qquad g_t = \frac{G_t}{c_t}.$$

Thus $Z_T = Z$, $q_{t-1}(G_t) = c_t$, and, for every function f,

$$q_{t-1}(g_t f) = q_t(f). \tag{1}$$

The empirical measure $\widehat{q}_t$ is taken after stage-t resampling and mutation. The particles start iid from q_0, with the exact normaliser $\widehat{Z}_0 = Z_0$. The evidence update and its normalised form are

$$\widehat{Z}_t = \widehat{Z}_{t-1}\widehat{q}_{t-1}(G_t), \qquad W_t := \frac{\widehat{Z}_t}{Z_t} = W_{t-1}\widehat{q}_{t-1}(g_t), \qquad W_0 = 1. \tag{2}$$

Each stage resamples multinomially and mutates with the method's q_t-invariant kernel K_t. All settings are fixed before final randomness (CAL), and all operations use exact arithmetic and fresh ideal streams (EX-A, EX-S). The four final islands use independent streams. No conditioning on final diagnostics is performed.

Theorem 1 (T25(iv), conditional form)**.** *For each $t = 1, \ldots, T$, let a_t, ν_t be finite nonnegative constants fixed given c. Suppose that the actual island satisfies*

$$\mathbb{E}_c\Big[W_{t-1}^2\{(\widehat{q}_{t-1}(g_t))^2 - 1\}\Big] \le \frac{a_t}{N}\mathbb{E}_c W_{t-1}^2 + \nu_t. \tag{PW2}$$

Then

$$\mathrm{Var}_c(W_T) \le B_T := \prod_{t=1}^{T}\left(1 + \frac{a_t}{N}\right) - 1 + \sum_{s=1}^{T}\nu_s \prod_{t=s+1}^{T}\left(1 + \frac{a_t}{N}\right). \tag{3}$$

If each of the four final islands satisfies PW2 with these same constants, then for $W_{\mathrm{pool}} = \frac{1}{4}\sum_{r=1}^{4} W_{r,T} = \widehat{Z}/Z$,

$$\mathrm{Var}_c(W_{\mathrm{pool}}) \le \frac{B_T}{4}. \tag{4}$$

Empty products are one. Here $N = N_e$ *throughout.*

2 Finiteness and the first moment

Every $d(S)$ is finite. Write $d_- = \min_\Omega d$ and $d_+ = \max_\Omega d$. Since the grid is increasing,

$$e^{\varphi_t d_-} \le \frac{\widehat{Z}_t}{Z_0} \le e^{\varphi_t d_+}.$$

Also $Z_t > 0$ is deterministic given c. Therefore W_t has finite moments of every order for each fixed c and t. In particular all expectations in PW2 and the following calculations are well defined.

Wave-1 T12(i.a), transferred to conditional calibration in T12(iii), gives $\mathbb{E}_c[\widehat{Z}_t \widehat{q}_t(f)] = Z_t q_t(f)$. For completeness, its needed first-moment argument follows directly from the prescribed operations. Let $\mathcal{F}_{t-1}$ include the island's history through stage $t-1$. Multinomial resampling and mutation give

$$\mathbb{E}_c[\widehat{q}_t(f) \mid \mathcal{F}_{t-1}] = \frac{\widehat{q}_{t-1}(G_t K_t f)}{\widehat{q}_{t-1}(G_t)}. \tag{5}$$

Indeed each child first chooses a parent with weights proportional to G_t and then moves with K_t; averaging its conditional expectation over the N children proves the identity.

We prove simultaneously for all functions $f : \Omega \to \mathbb{R}$ that

$$\mathbb{E}_c[W_t \widehat{q}_t(f)] = q_t(f). \tag{6}$$

At $t = 0$ this follows from iid q_0 starts and $W_0 = 1$. For the induction step, W_t in (2) is $\mathcal{F}_{t-1}$-measurable. Multiplying (5) by it, taking expectation and using the induction hypothesis with the deterministic function $g_t K_t f$ gives

$$\begin{aligned}\mathbb{E}_c[W_t \widehat{q}_t(f)] &= \mathbb{E}_c[W_{t-1} \widehat{q}_{t-1}(g_t K_t f)] \\ &= q_{t-1}(g_t K_t f) = q_t(K_t f) = q_t(f).\end{aligned}$$

The last two equalities use (1) and invariance. With $f \equiv 1$ this yields

$$\mathbb{E}_c W_t = 1, \qquad \mathrm{Var}_c(W_t) = \mathbb{E}_c W_t^2 - 1. \tag{7}$$

This is an unconditional first moment under the frozen-run law; it does not assert $\mathbb{E}_c[W_t \mid \mathcal{F}_{t-1}] = W_{t-1}$.

3 Second-moment recursion and its solution

Set $M_t = \mathbb{E}_c W_t^2$ and $\alpha_t = 1 + a_t/N \ge 1$. Squaring the pathwise update (2) and subtracting W_{t-1}^2 gives the exact identity

$$M_t - M_{t-1} = \mathbb{E}_c\Big[W_{t-1}^2 \{(\widehat{q}_{t-1}(g_t))^2 - 1\}\Big]. \tag{8}$$

Applying PW2 to this identity yields

$$M_t \le \alpha_t M_{t-1} + \nu_t, \qquad M_0 = 1. \tag{9}$$

By (8), PW2 is equivalent to the recursion $M_t \le (1 + a_t/N)M_{t-1} + \nu_t$. The theorem therefore adds no control of the evidence beyond PW2 itself; all of its substance lies in establishing PW2 with useful constants, which is not done here. Define

$$U_0 = 1, \qquad U_t = \prod_{j=1}^{t} \alpha_j + \sum_{s=1}^{t} \nu_s \prod_{j=s+1}^{t} \alpha_j.$$

Separating the last term of the sum shows $U_t = \alpha_t U_{t-1} + \nu_t$. Induction now proves $M_t \le U_t$: the base case is equality, and if $M_{t-1} \le U_{t-1}$ then

$$M_t \le \alpha_t M_{t-1} + \nu_t \le \alpha_t U_{t-1} + \nu_t = U_t,$$

where the second inequality uses $\alpha_t \ge 0$. By (7), $\mathrm{Var}_c(W_T) = M_T - 1 \le U_T - 1 = B_T$. This proves (3), with the slack at stage s multiplied only by the factors from later stages $t > s$.

4 Four independent islands

Under CAL and EX-S, conditional on c, each island is a measurable function of the fixed settings and its own fresh ideal stream. Thus the four evidence ratios are independent and identically distributed given c, as also established in wave-1 T12(iv.c). Each has mean one by (7), and each has variance at most B_T. Hence

$$\mathrm{Var}_c\left(\frac{1}{4}\sum_{r=1}^{4} W_{r,T}\right) = \frac{1}{16}\sum_{r=1}^{4} \mathrm{Var}_c(W_{r,T}) \le \frac{4B_T}{16} = \frac{B_T}{4}.$$

The factor $1/4$ applies to both terms in (3). It is not obtained by replacing N_e with $4N_e$ inside that formula. Terminal rounds and local search leave the evidence unchanged, so this is the bound for the reported pooled evidence. Since the argument holds for each successful frozen calibration, it holds almost surely given $\mathcal{C}$ on calibration success. This completes the proof. □

Dependency record. The wave-1 source is `COLLECTED_PROOFS_v2.tex`, chapter "T12 (T8): frozen-run evidence and recorded Rao–Blackwell identities", parts (i.a), (iii), (iv.c). These are used as proved, not adopted results; the needed first-moment induction is also displayed above. No unproved v3 conjecture is invoked. In particular, PW2 remains a hypothesis.